received and processed by the frontend to update the chat interface and display the LLM's generated response to the user.

- *Error handling*—The backend incorporates error-handling mechanisms to catch unforeseen problems. In case of errors (e.g., invalid requests, API errors), the backend responds with appropriate status codes to inform the frontend.

A visual representation of the backend layout is depicted in figure 2.8, providing a clear overview of the backend server's functionalities and its interaction with the OpenAI service.

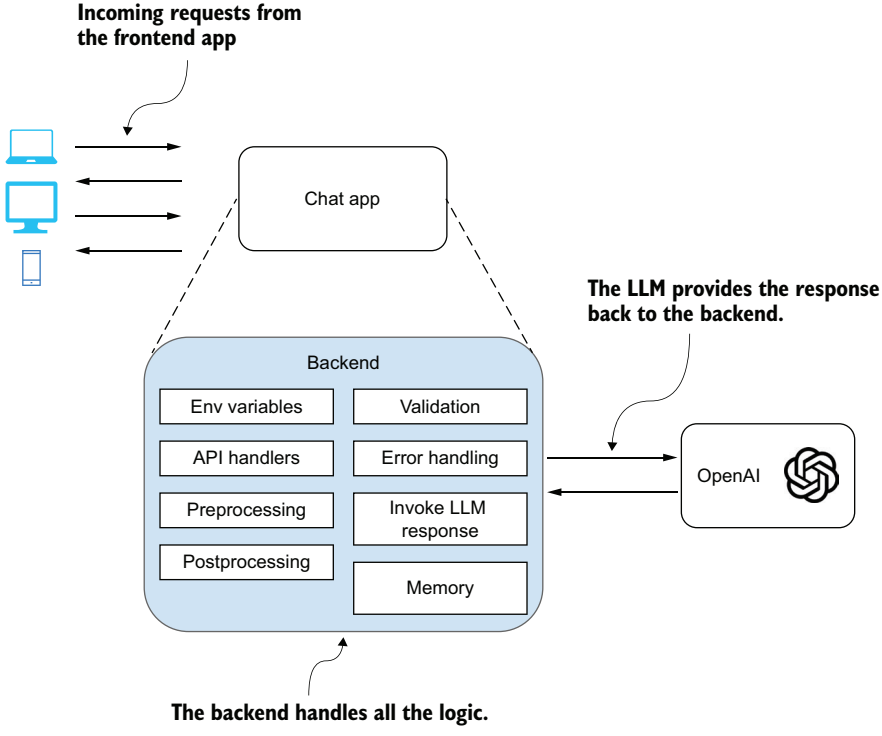


Figure 2.8 The general responsibilities of the backend. The process involves data validation, interaction with the OpenAI API, response processing, and transmission of the generated response back to the frontend application.

Now we will look at the code that powers the OpenAI backend. Listing 2.4 provides a closer look at the code that sets up the server (server.openai.js).

> **NOTE** The code that powers the Google Gemini backend is located in a separate file named server.google.js. This file is almost identical to the OpenAI backend implementation, with the primary difference being the calls made to the respective client libraries for each service.